

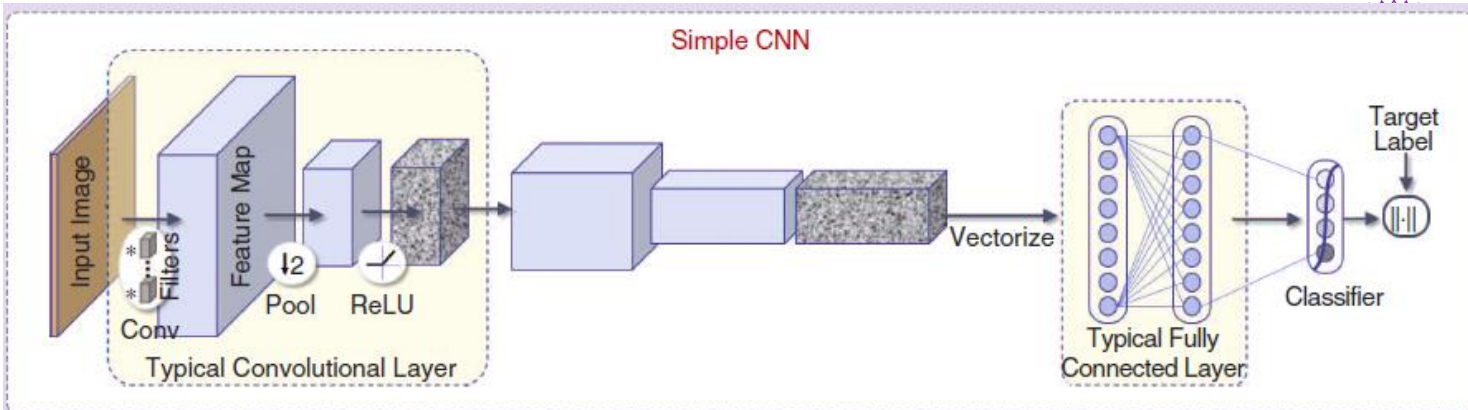


CNN-Based Image Restoration

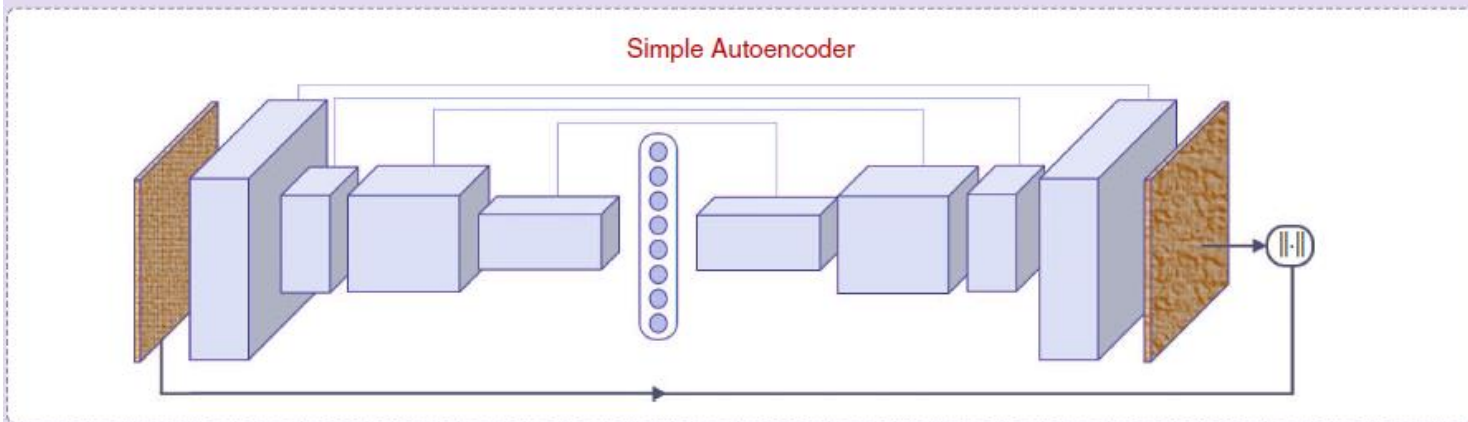
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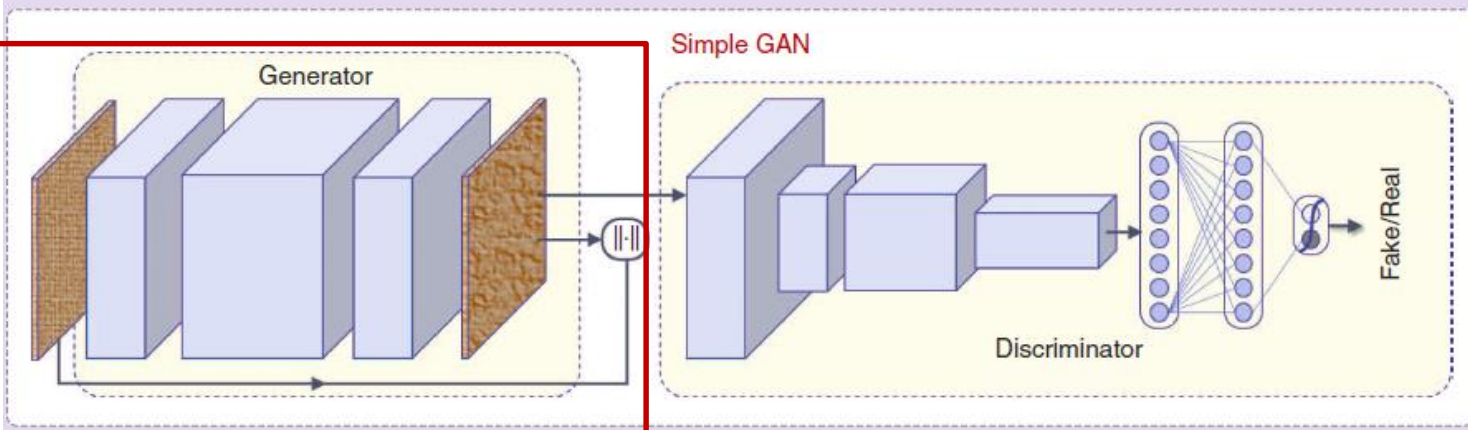
Extract low-dim features from image (high-dim),
e.g. classification



Extract low-dim features first and then build dense label maps,
e.g. segmentation, deblur (U-Net)



Generate locally fine texture (globally high-dim),
e.g. image restoration



Credit: F. Porikli et al, *IEEE Signal Processing Magazine*, 2018.



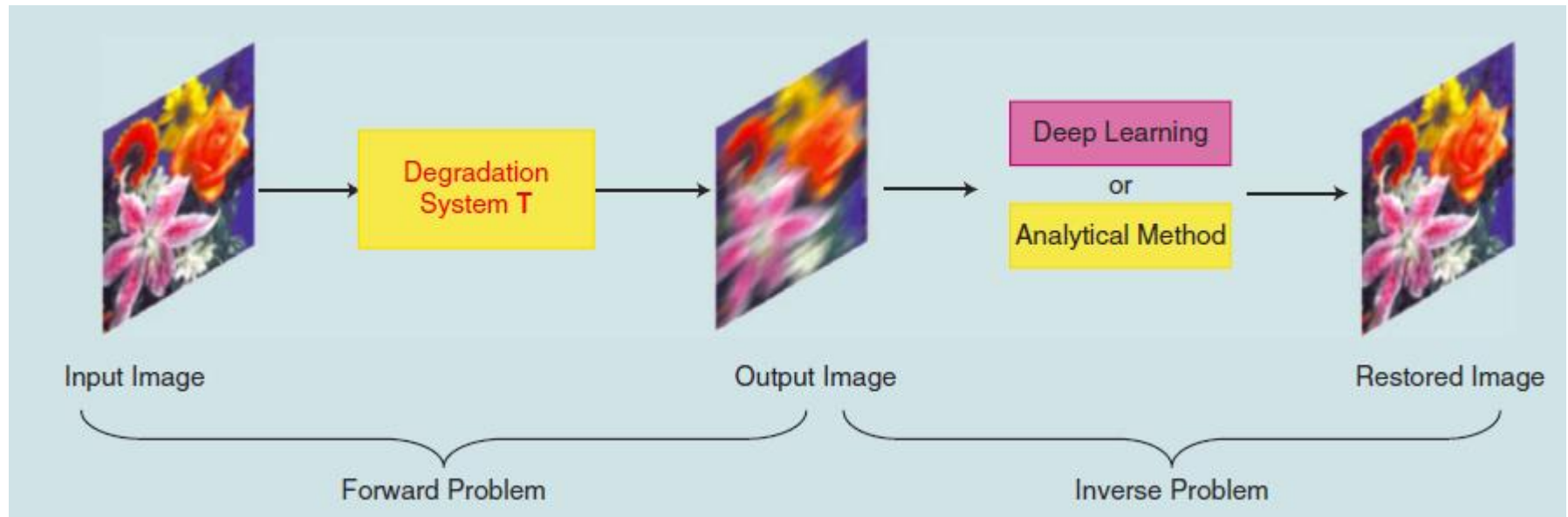
Outline

- **CNN-based image restoration**
- CNN-based super-resolution

Ref: A. Lucas et al, "Using Deep Neural Networks for Inverse Problems in Imaging", *IEEE Signal Processing Magazine*, 2018.

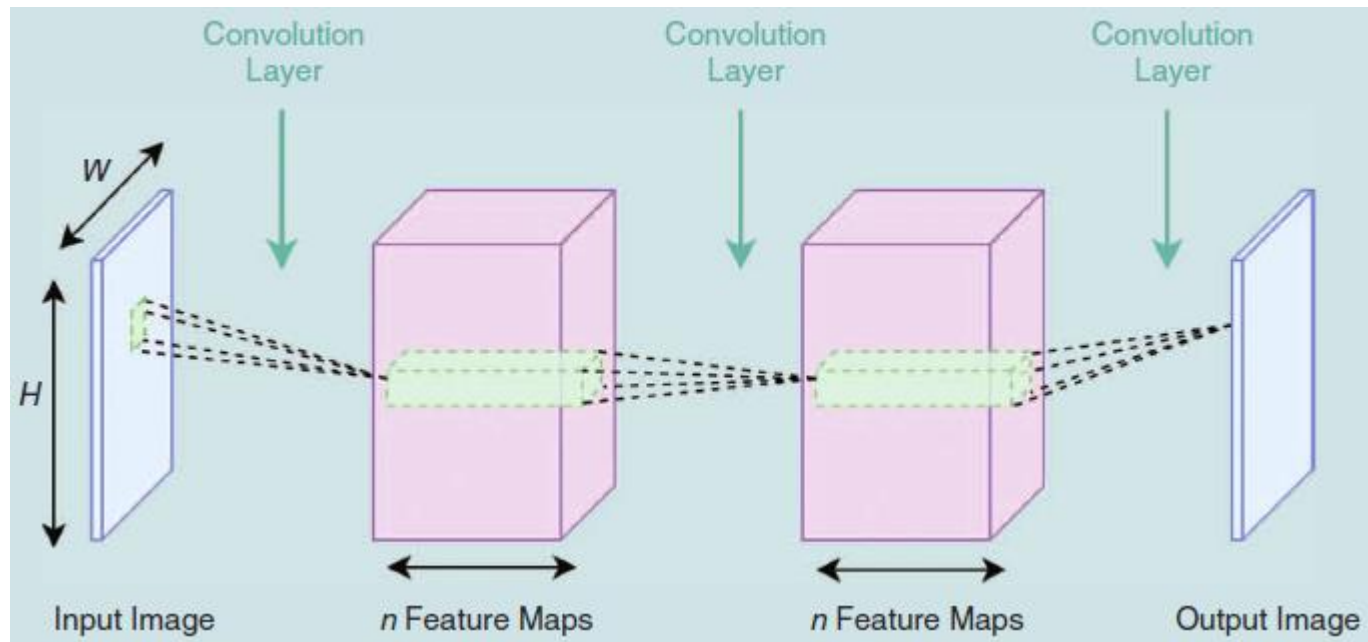
Image restoration as inverse problem

- Including denoising, deblurring, super-resolution, inpainting, ...
 - In a more general sense, also including motion estimation, depth estimation, ... (find hidden information)



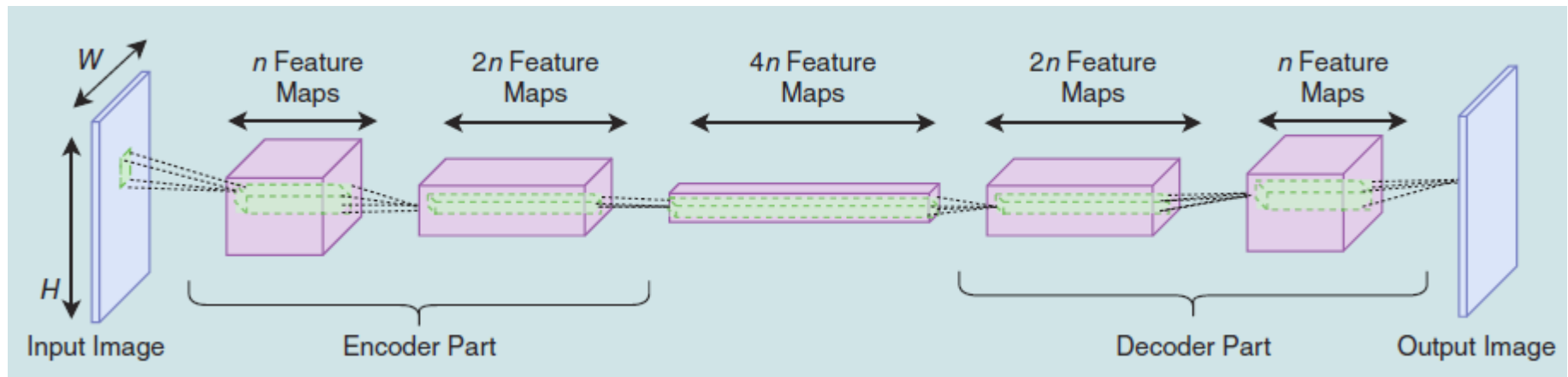
Generative network

- Fully convolutional network with zero padding
 - Linear output for image restoration problems
 - Non-linear activation after convolution
 - Dense-to-dense pixel generation



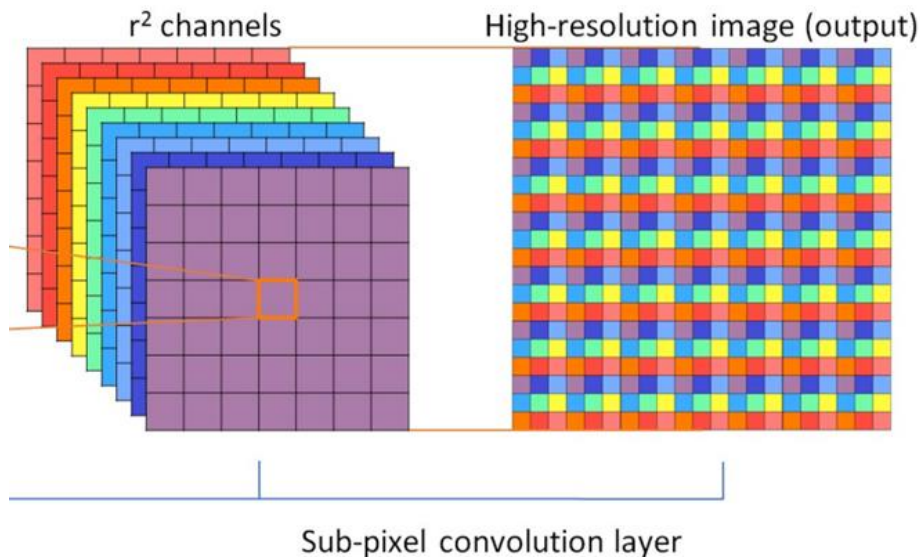
Auto-encoder

- Encoder-decoder architecture
 - For image segmentation problems
 - Encoder spatially compresses features
 - Decoder increases back the spatial size
 - U-Net: Encoder-decoder + many skip connections

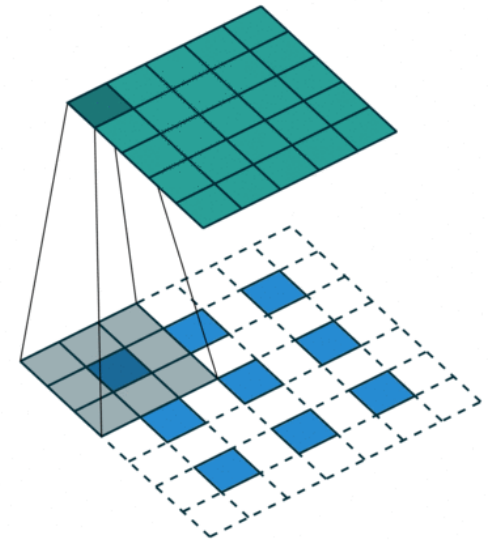


Upsample layer

- Transposed convolution or fractionally strided convolution
- Sub-pixel convolution (pixel shuffle)



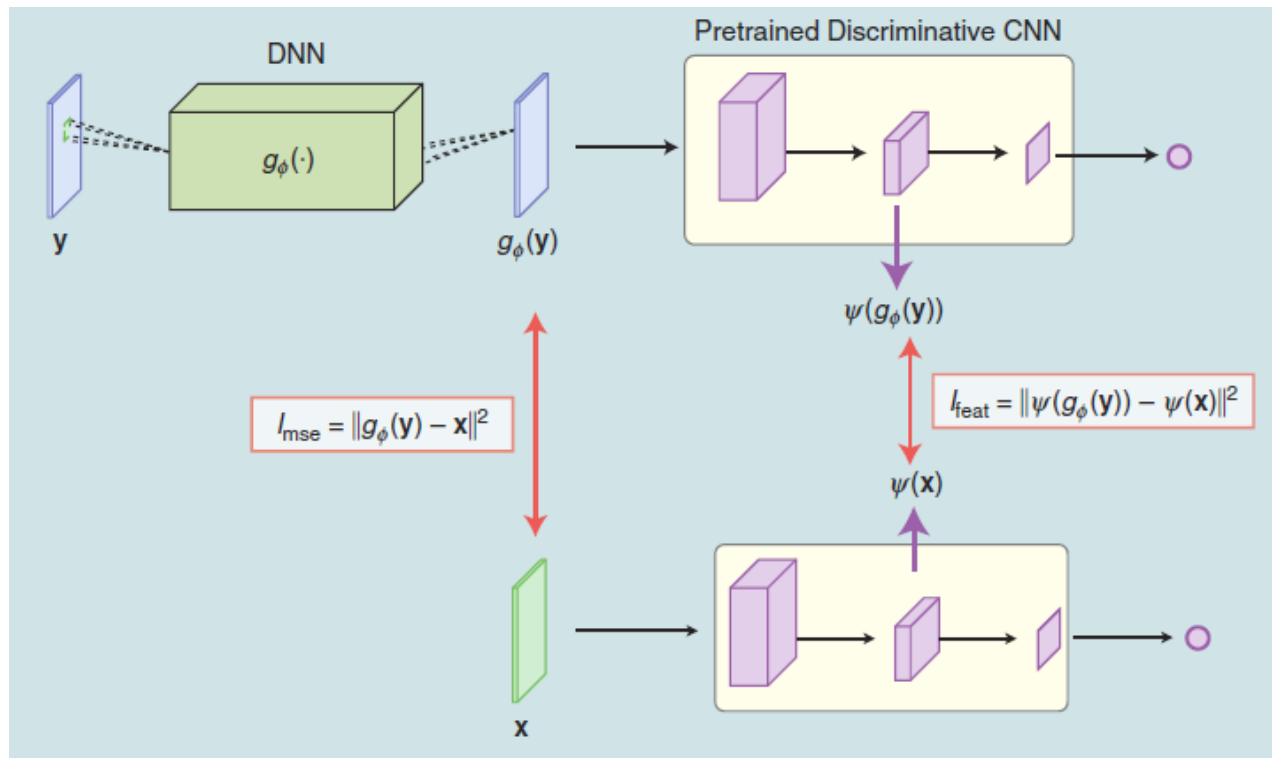
Ref: <https://arxiv.org/pdf/1609.05158.pdf>



Ref: https://github.com/vdumoulin/conv_arithmetic/

Training loss function

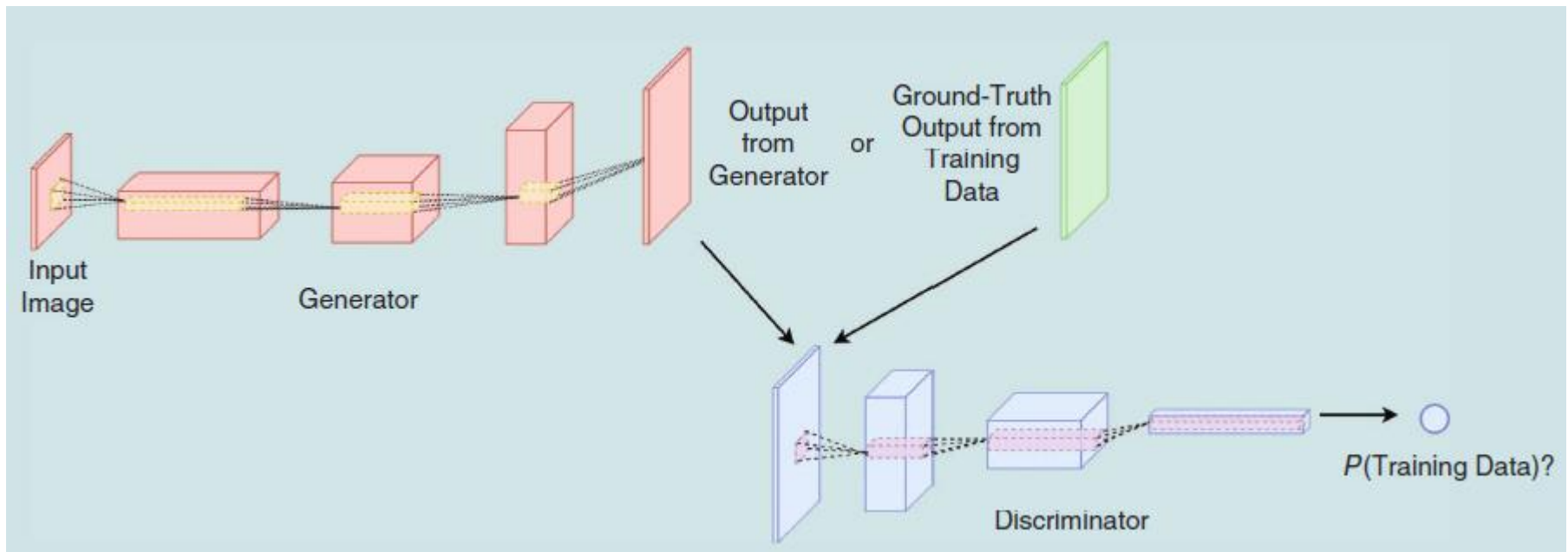
- Analytical metric (MSE, L1, ...)
- Pre-trained feature distance (VGGNet, ...)



- Generative adversarial network (GAN)

GAN framework

- Introduce a discriminator network to make generator more realistic
 - Can combine analytical or pre-trained feature metrics



Limitation of learning-based method

- Reliability problem: can't handle unseen cases
 - Problems, if any, are harder to trace (explainability, accountability)



- Better than analytical methods on average; but could be worse in extreme cases
- Incorporating analytical priors becomes indirect
 - e.g. via data augmentation



Note on network differences

Classification task	Restoration task
On small interesting image blocks Many crops may be tested	On full-resolution images High complexity (fully convolutional network)
Capture global features - Prefer intensive pooling and striding - Quality ↗, as batch size ↗	Capture local features - Prefer less pooling and striding - Quality ↘, as batch size ↗
Should resist local variance Prefer weight perturbation	Should adapt to local variance Prefer less perturbation, e.g. no BN
Output layer inflates features - Should avoid exploding weights - Prefer BN and weight decay	Output layer confines features - Should leave freedom for weights - Prefer no BN and no weight decay
Lots of redundancy in parameters Compression hurts quality gracefully	Less redundancy in parameters Compression hurts quality directly

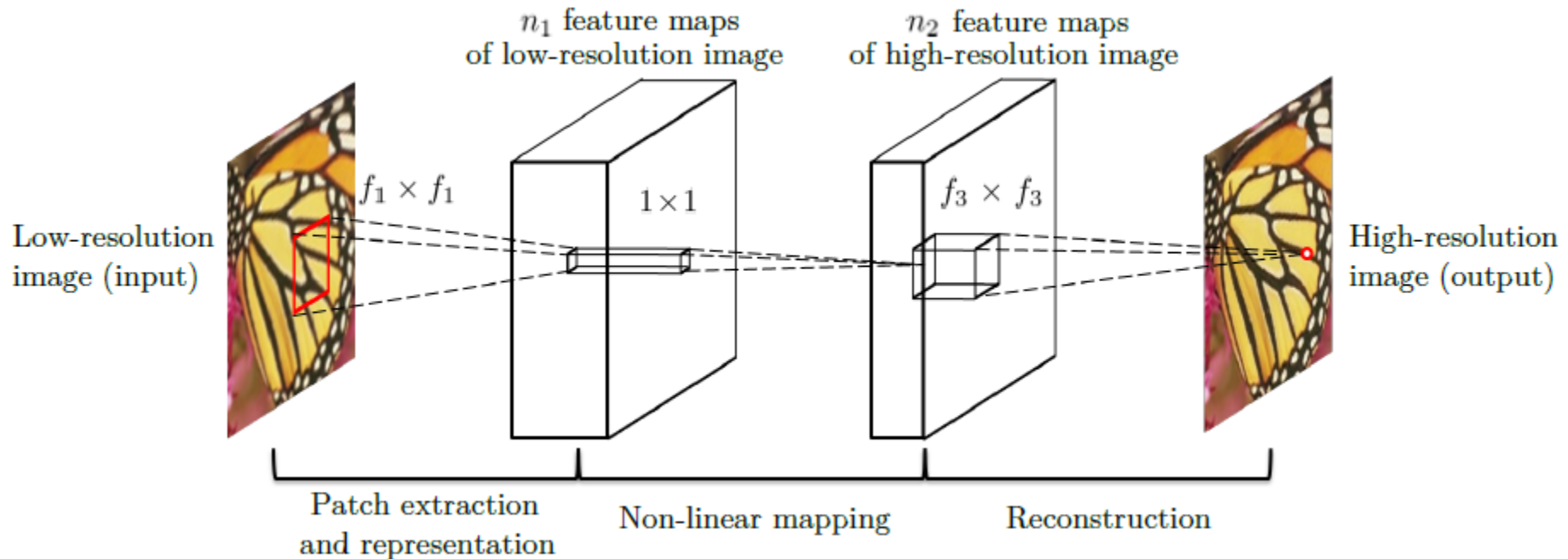
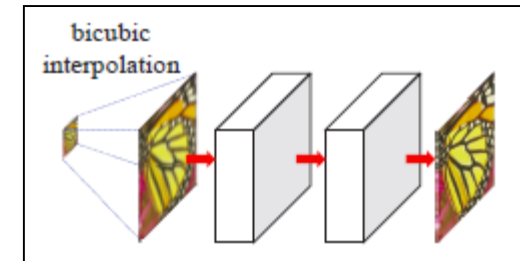


Outline

- CNN-based image restoration
- **CNN-based super-resolution**

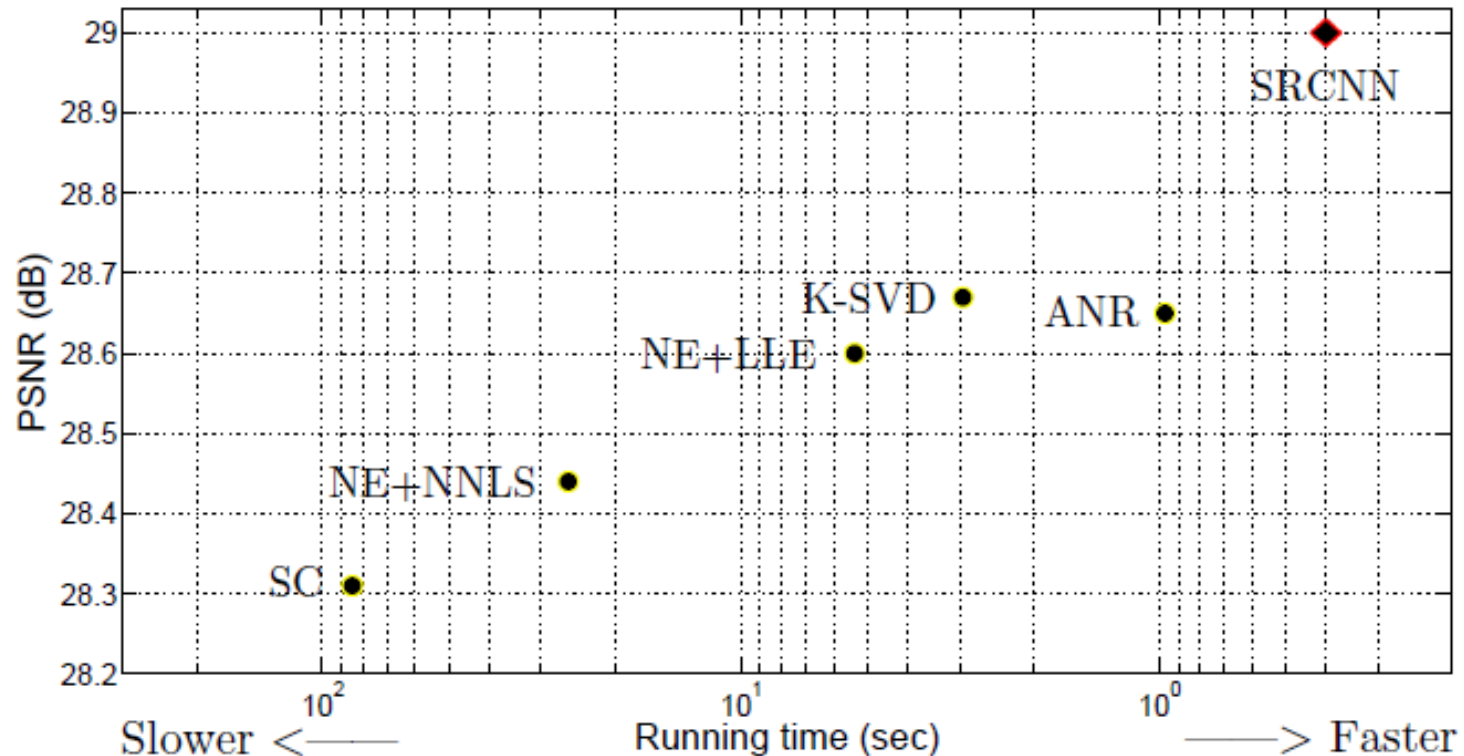
SRCNN [ECCV'14]

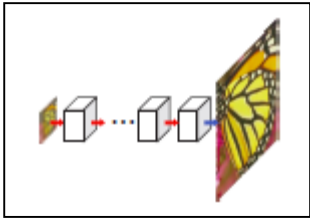
- Three-layer network
 - CNN on upsampled images



SRCNN result

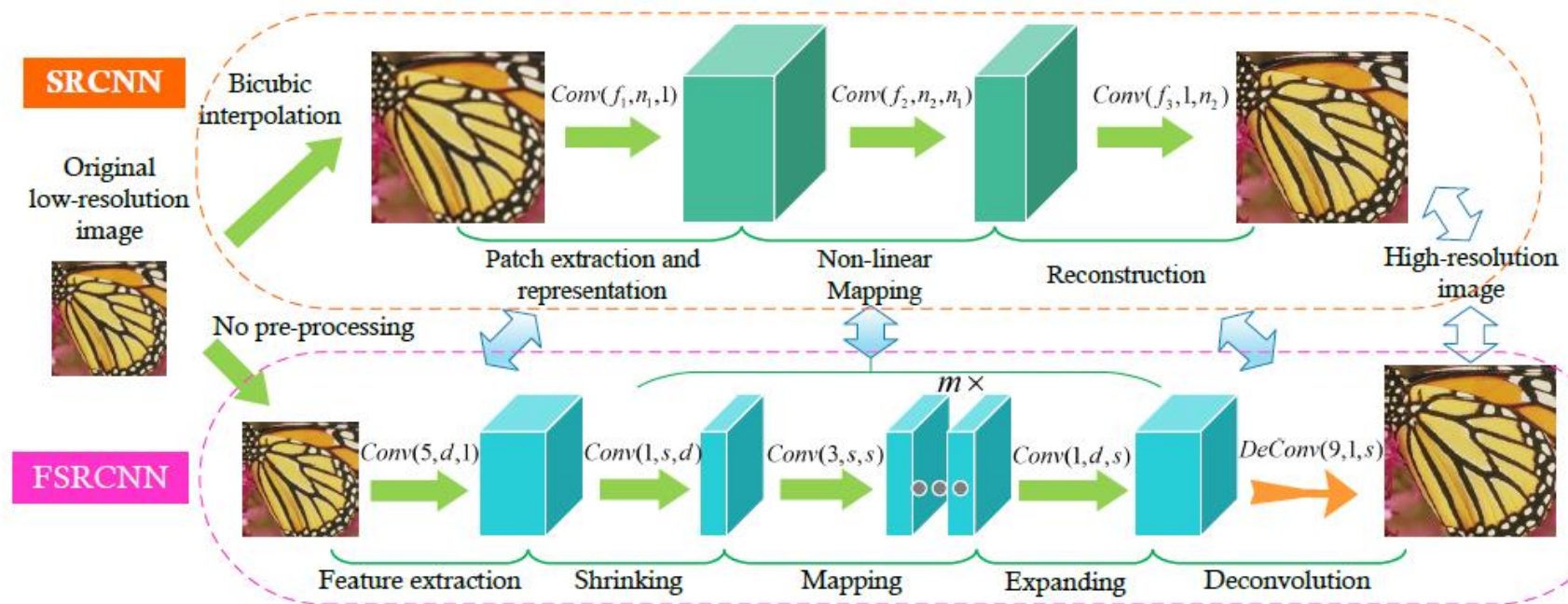
- Better in terms of quality and speed compared to previous work (x3 scale)





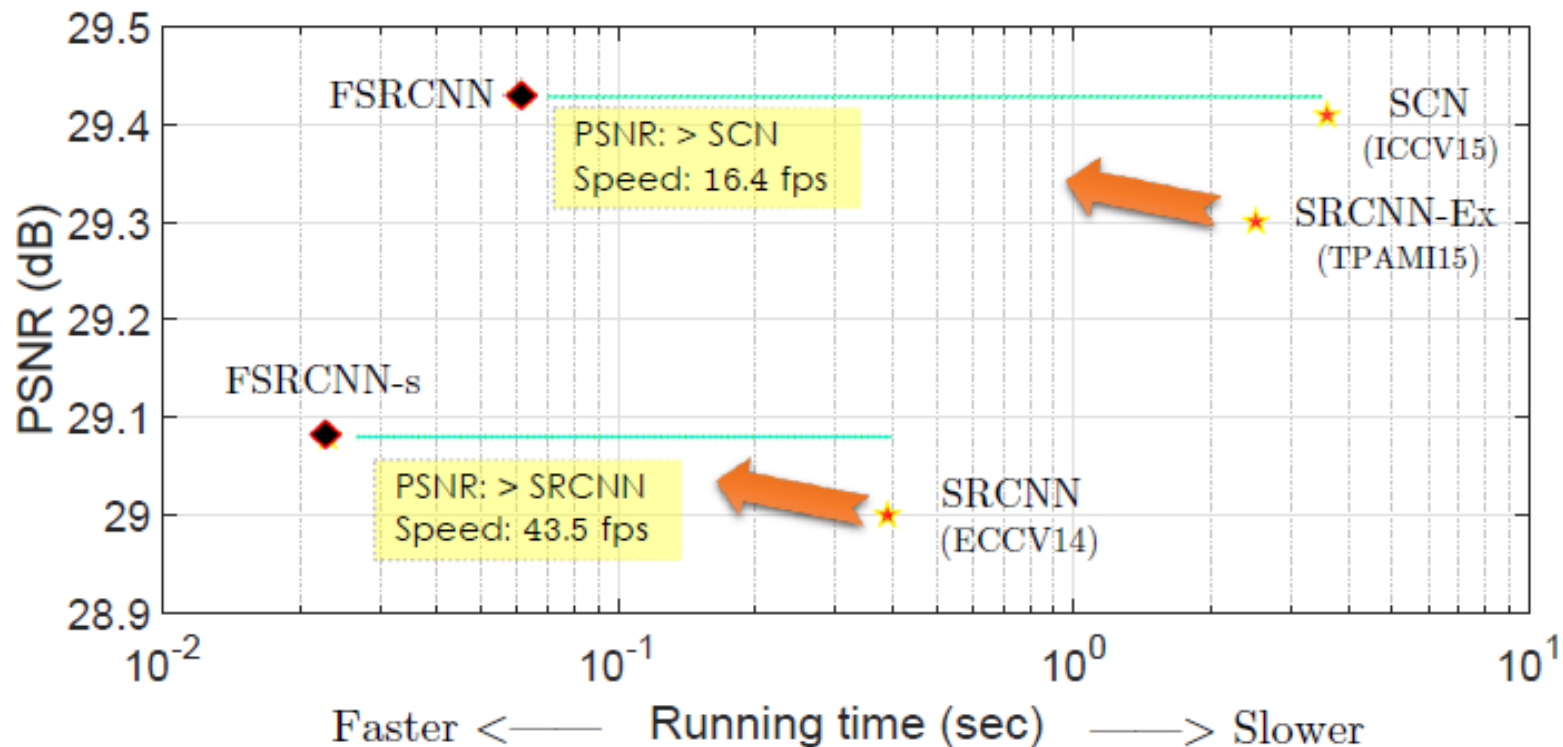
FSRCNN [ECCV'16]

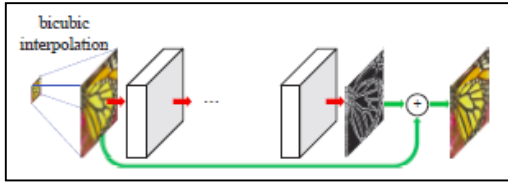
- Use fractionally strided convolution for upsampling at the end of the network to reduce complexity



FSRCNN result

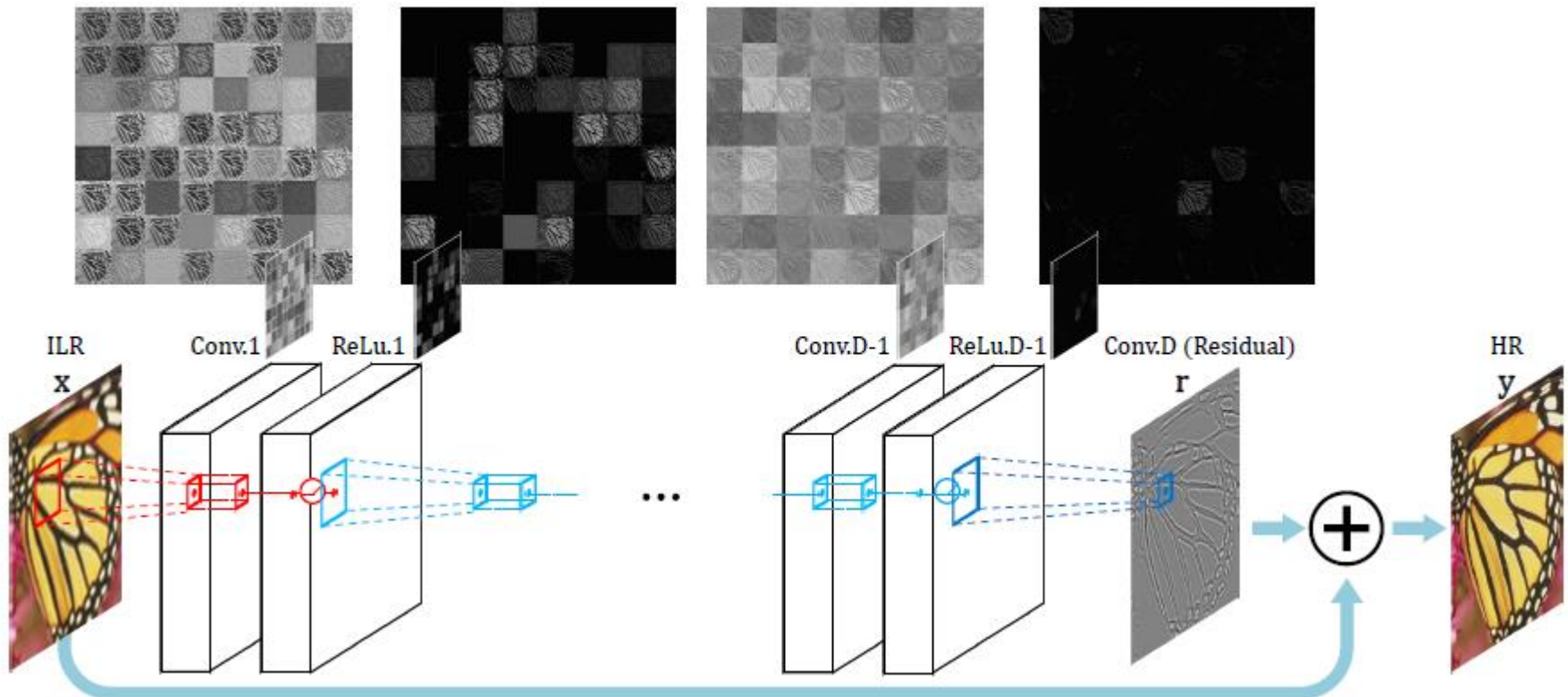
- Better in terms of quality and speed compared to SRCNN (x3 scale)



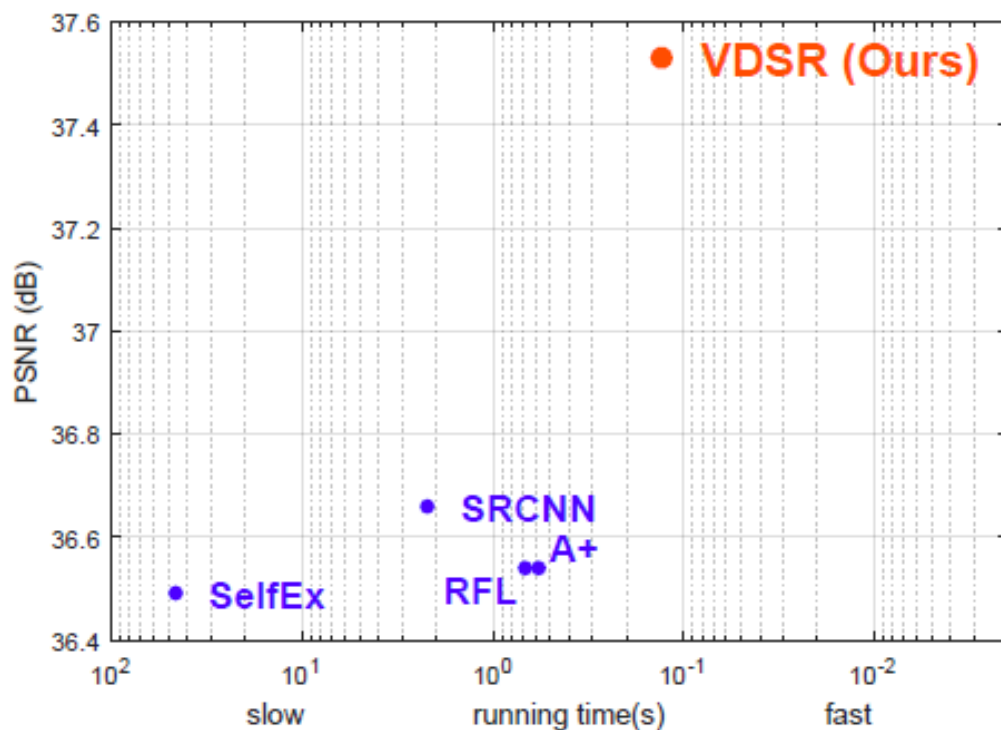


VDSR [CVPR'16]

- Skip connection for learning residuals
- Deep layers (20) to improve quality



VDSR result



Epoch	10	20	40	80
Residual	36.90	36.64	37.12	37.05
Non-Residual	27.42	19.59	31.38	35.66
Difference	9.48	17.05	5.74	1.39

(a) Initial learning rate 0.1

Epoch	10	20	40	80
Residual	36.74	36.87	36.91	36.93
Non-Residual	30.33	33.59	36.26	36.42
Difference	6.41	3.28	0.65	0.52

(b) Initial learning rate 0.01

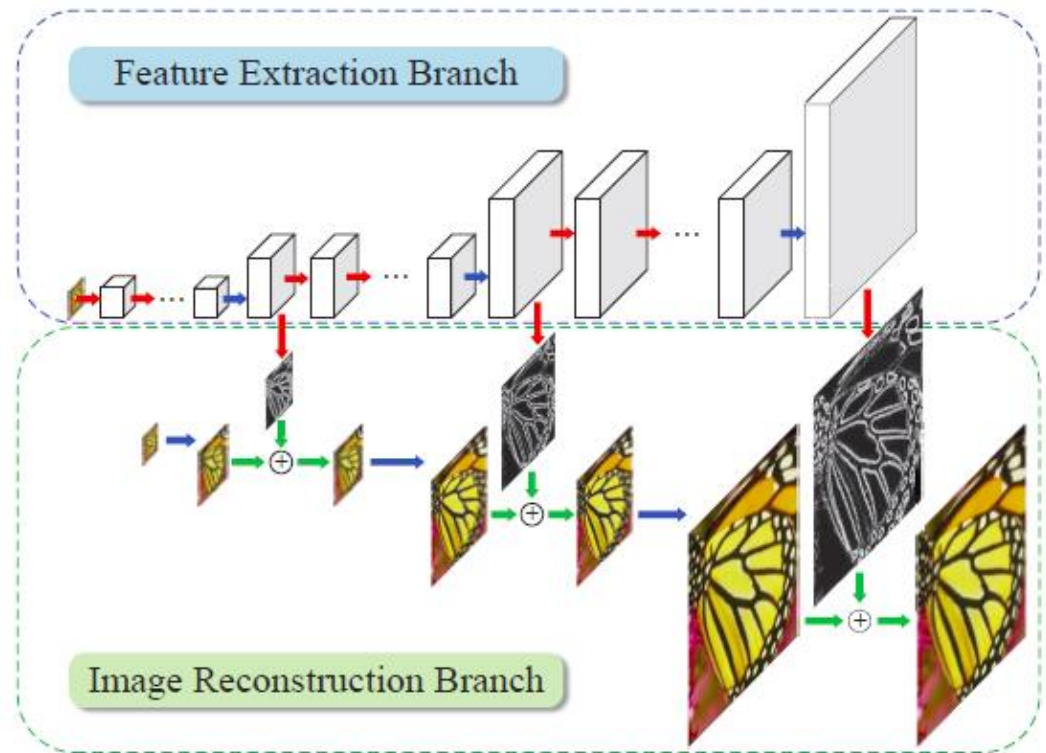
Epoch	10	20	40	80
Residual	36.31	36.46	36.52	36.52
Non-Residual	33.97	35.08	36.11	36.11
Difference	2.35	1.38	0.42	0.40

(c) Initial learning rate 0.001

LapSRN [CVPR'17]

- Laplacian pyramid to update details progressively and generate multi-scale (2x/4x/8x) images in one pass
 - Residual learning
- Deeper layers (27) to improve quality
- Robust Charbonnier cost (Differentiable variant of L1)

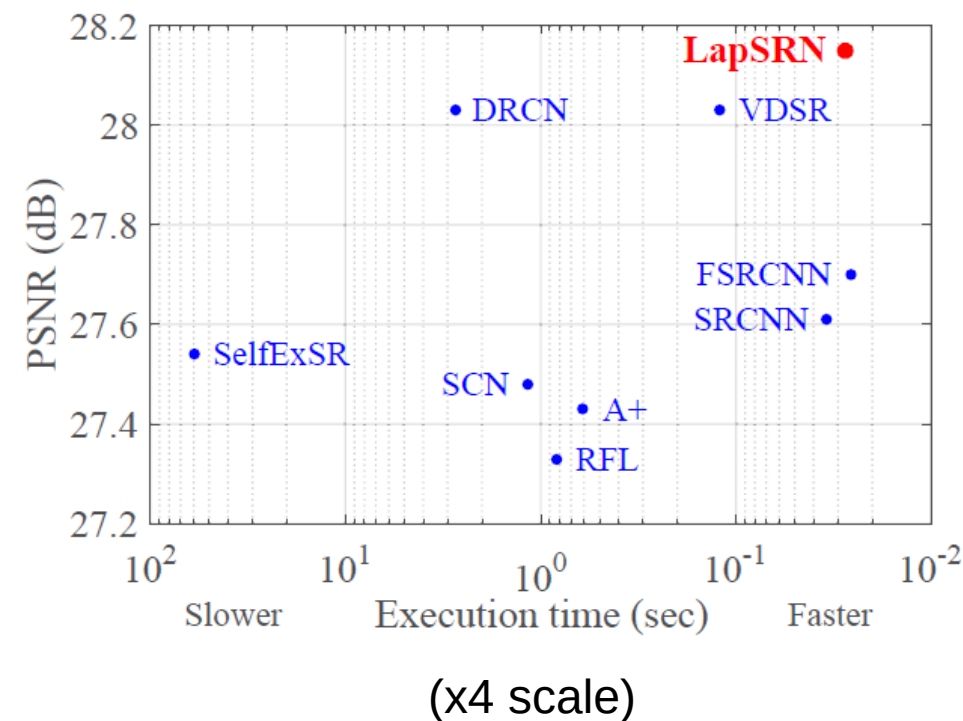
$$\rho(x) = \sqrt{x^2 + \varepsilon^2}$$



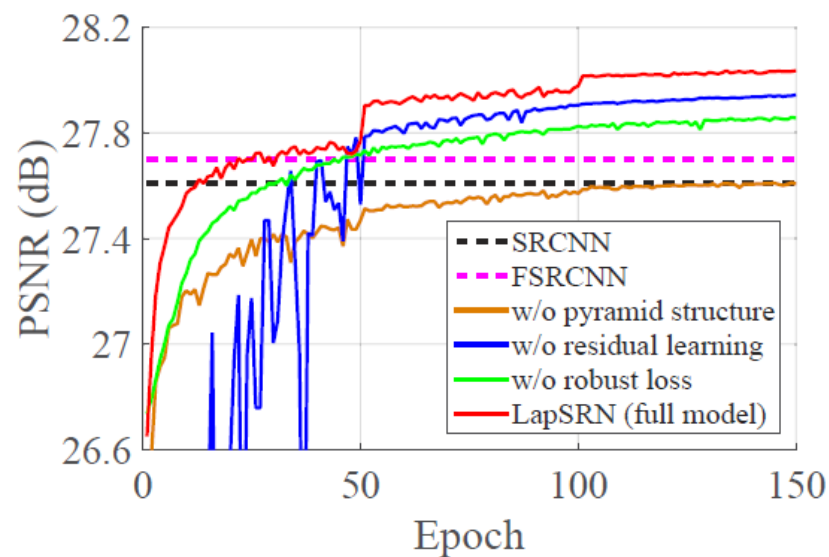
Red: convolution

Blue: transposed convolution

LapSRN result

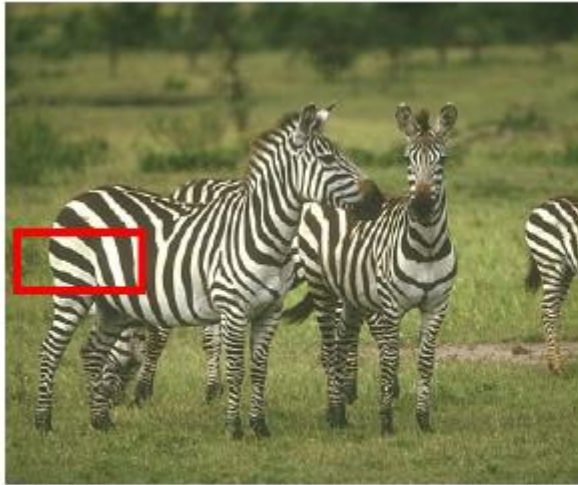


Ablation study



Residual	Pyramid	Loss	SET5	SET14
✓		Robust	30.58	27.61
	✓	Robust	31.10	27.94
✓	✓	ℓ_2	30.93	27.86
✓	✓	Robust	31.28	28.04

LapSRN result



Ground-truth HR



HR
(PSNR, SSIM)



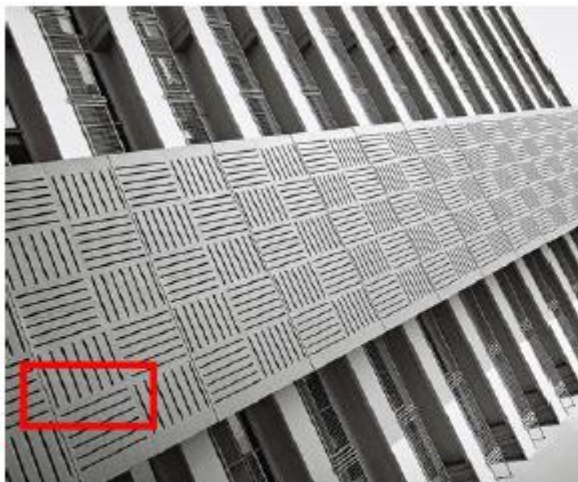
FSRCNN [8]
(19.57, 0.5133)



VDSR [17]
(19.58, 0.5147)



LapSRN (ours)
(19.75, 0.5246)



Ground-truth HR



HR
(PSNR, SSIM)



FSRCNN [8]
(15.61, 0.3494)



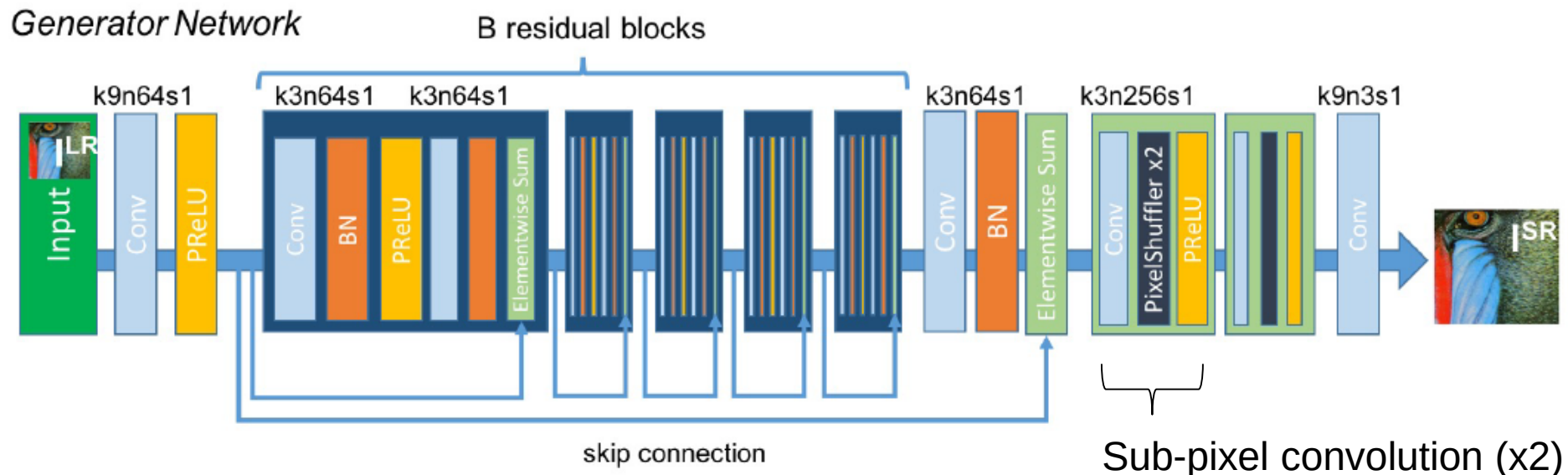
VDSR [17]
(15.66, 0.3644)



LapSRN (ours)
(15.72, 0.3865)

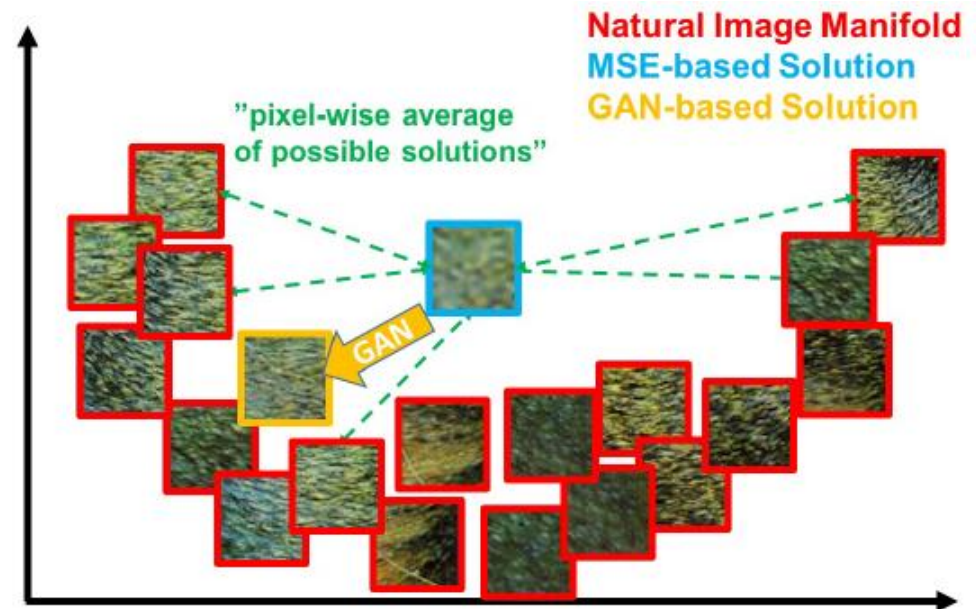
SRResNet [CVPR'17]

- Residual blocks to stabilize training procedure
- Deeper layers (37, B=16) to improve quality
- VGGNet feature cost to improve subjective quality

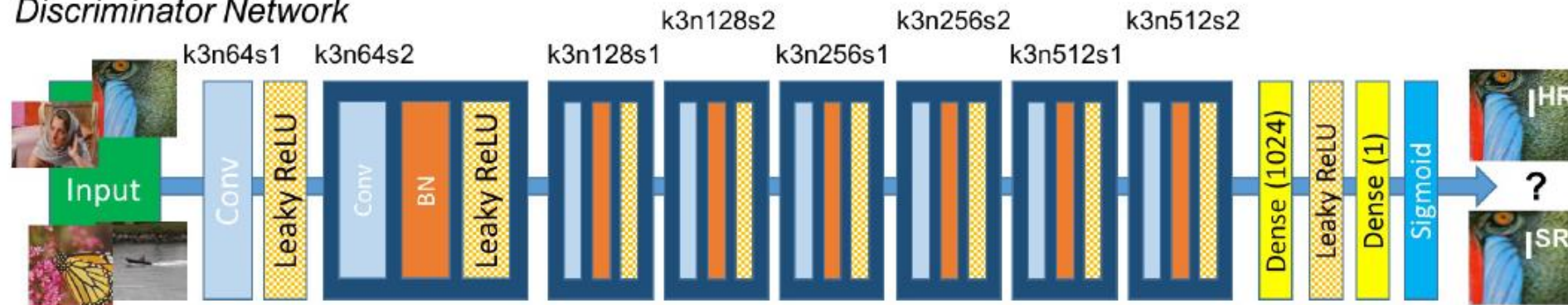


SRGAN

- Use GAN to generate realistic photos

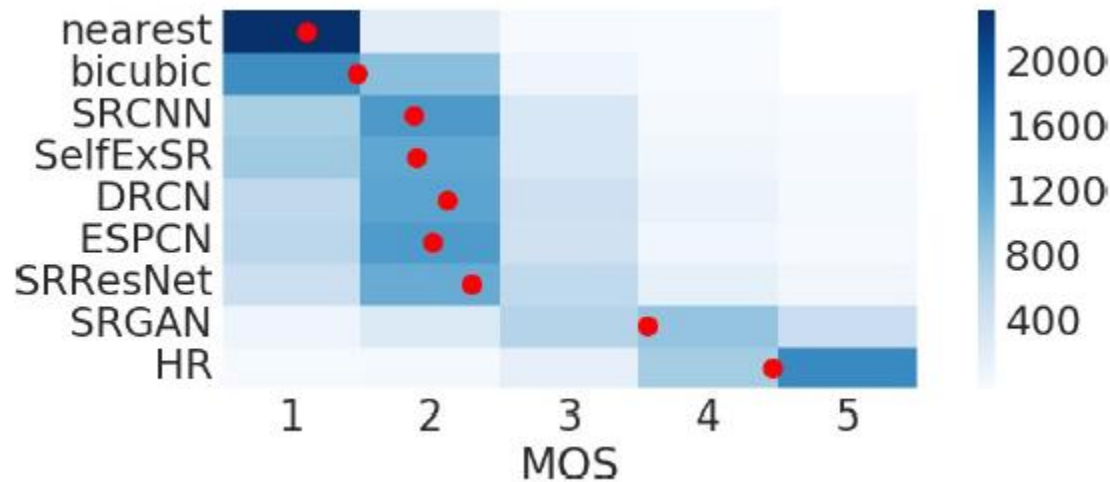


Discriminator Network



SRResNet/SRGAN result

Set5	SRResNet-			SRGAN-	
	MSE	VGG22		MSE	VGG22 VGG54
PSNR	32.05	30.51		30.64	29.84 29.40
SSIM	0.9019	0.8803		0.8701	0.8468 0.8472
MOS	3.37	3.46		3.77	3.78 3.58
Set14					
PSNR	28.49	27.19		26.92	26.44 26.02
SSIM	0.8184	0.7807		0.7611	0.7518 0.7397
MOS	2.98	3.15*		3.43	3.57 3.72*

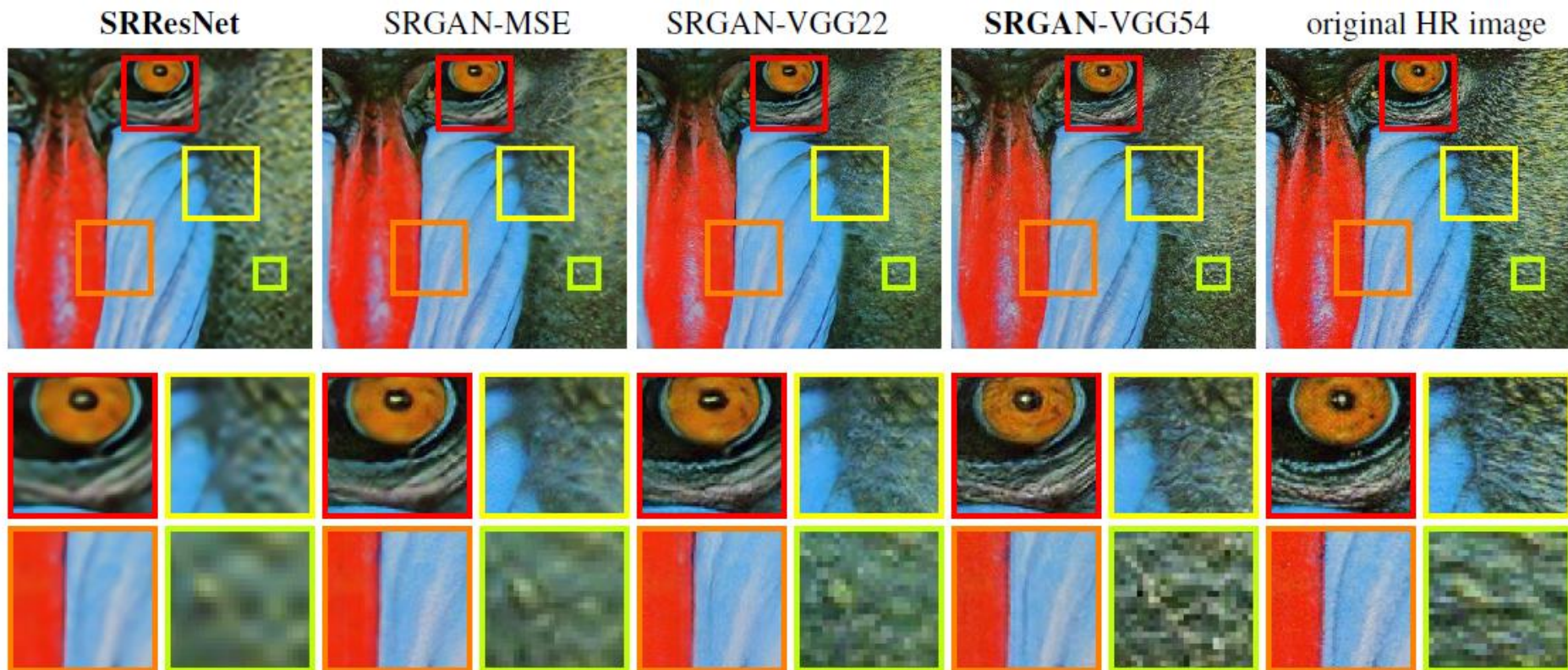




SRResNet/SRGAN result

Set5	nearest	bicubic	SRCNN	SelfExSR	DRCN	ESPCN	SRResNet	SRGAN	HR
PSNR	26.26	28.43	30.07	30.33	31.52	30.76	32.05	29.40	∞
SSIM	0.7552	0.8211	0.8627	0.872	0.8938	0.8784	0.9019	0.8472	1
MOS	1.28	1.97	2.57	2.65	3.26	2.89	3.37	3.58	4.32
Set14									
PSNR	24.64	25.99	27.18	27.45	28.02	27.66	28.49	26.02	∞
SSIM	0.7100	0.7486	0.7861	0.7972	0.8074	0.8004	0.8184	0.7397	1
MOS	1.20	1.80	2.26	2.34	2.84	2.52	2.98	3.72	4.32
BSD100									
PSNR	25.02	25.94	26.68	26.83	27.21	27.02	27.58	25.16	∞
SSIM	0.6606	0.6935	0.7291	0.7387	0.7493	0.7442	0.7620	0.6688	1
MOS	1.11	1.47	1.87	1.89	2.12	2.01	2.29	3.56	4.46

SRResNet/SRGAN result



SRResNet/SRGAN result

bicubic
(21.59dB/0.6423)



SRResNet
(23.53dB/0.7832)



SRGAN
(21.15dB/0.6868)



original



Failure case



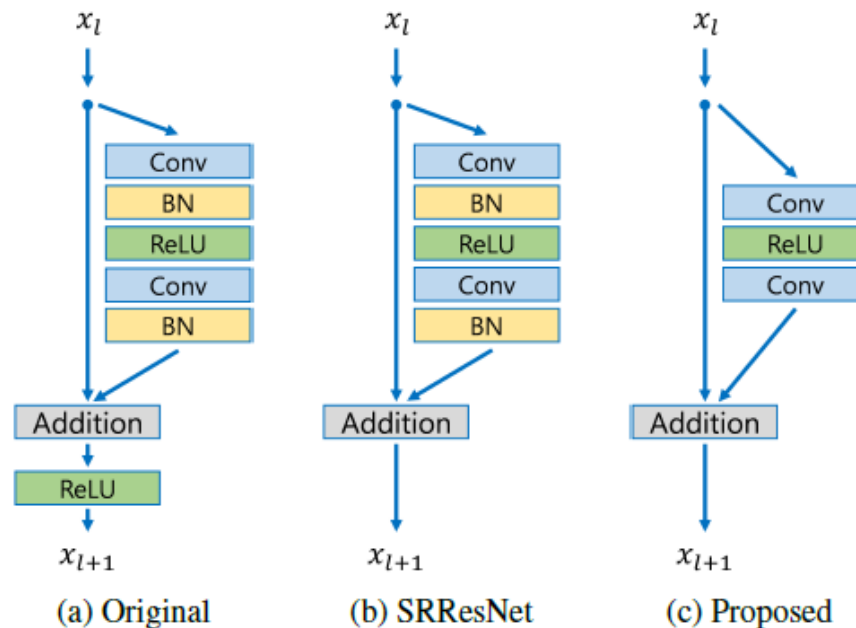
SRGAN-VGG54



SRResNet-MSE

EDSR [CVPRW'17]

- Enhance SRResNet structure by removing BN layers and some ReLUs
- Improve quality by increasing depth and feature number
- Improve quality by self-ensemble method





EDSR result

B=16, F=64

B=32, F=128

Scale	SRResNet (L2 loss)	SRResNet (L1 loss)	Our baseline (Single-scale)	Our baseline (Multi-scale)	EDSR (Ours)	MDSR (Ours)	EDSR+ (Ours)	MDSR+ (Ours)
×2	34.40 / 0.9662	34.44 / 0.9665	34.55 / 0.9671	34.60 / 0.9673	35.03 / 0.9695	34.96 / 0.9692	35.12 / 0.9699	35.05 / 0.9696
×3	30.82 / 0.9288	30.85 / 0.9292	30.90 / 0.9298	30.91 / 0.9298	31.26 / 0.9340	31.25 / 0.9338	31.39 / 0.9351	31.36 / 0.9346
×4	28.92 / 0.8960	28.92 / 0.8961	28.94 / 0.8963	28.95 / 0.8962	29.25 / 0.9017	29.26 / 0.9016	29.38 / 0.9032	29.36 / 0.9029

L1 loss
performs
similarly to L2

Removing BN
improves
quality

PSNR keeps
improving for
larger networks

Self-ensemble
improves PSNR
by ~0.1dB